No View from Nowhere

Read-out Closure and the Vacuum Curvature Test in Q-RAIF A

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Abstract

A source can be physical and still be missed by a restricted interface. For a finite-dimensional readout channel Λ and source observable H, the operator-norm distance from H to the dual range equals half the largest energy difference between states with the same output. It is also the smallest worst-case error of any energy estimate based on the exact output state, including nonlinear estimates. An actual finite instrument adds a separate sampling cost. We exhibit a positive two-part source whose hidden interaction gives a nonzero error floor, then remove that floor by adding a priced mixed-memory record of the same source.

Complete null stress data have a metric-trace blind direction. A real vacuum-density difference therefore cannot be dismissed as a zero-point convention merely because null readout misses it. A matched Gaussian material supplies a complementary test: changing its contact changes the fluctuation operator and its logarithmic vacuum coefficient, with one physical calibration required across configurations.

These results turn readout visibility into a source-certification problem. They support, without proving, a broader closure conjecture: a spacetime description must specify or generate the interfaces that select its records and sources. Whether those interfaces can be fully internalized is an independent question. Neither interface blindness nor the closure conjecture computes the cosmological constant or switches off physical vacuum gravitation.

1 What the reader leaves outside

A laboratory description places its detectors inside spacetime and lets their records describe the rest of the world. It also needs a rule for what counts as a detector, a record and a relevant source. The question behind "no view from nowhere" is whether those rules can themselves be generated and represented inside the same description, with their physical execution accounted for.

This is a question about structural interfaces, not a claim that a human mind creates the world. Q-RAIF A obtains a Lorentzian propagation form under a common quadratic-shell condition and its independence assumptions; the dynamics need not select one readout cone [20]. The construction makes the selection problem concrete. It does not prove that an observer-independent formulation is impossible. HAFF G's finite programming obstruction likewise

distinguishes a programmed target operation from its joint executor; it does not prohibit stored self-descriptions or typed fixed points.

Vacuum energy supplies a particularly demanding test. A description may know how to read radiation while lacking a trace-sensitive source datum. Calling the missing datum invisible does not settle its gravitational effect. We first determine exactly what a finite interface can certify, including how much source can remain hidden. We then apply that distinction to null stress, physical vacuum differences and changes of apparatus. The closure question returns after these concrete results.

2 A source certificate for a finite interface

Let Λ be a trace-preserving quantum channel between finite-dimensional matrix algebras and $H = H^*$ a specified source observable. The source datum is $\operatorname{tr} H\rho$. It may be an energy or a bounded smeared source in a regulated model. Its definition precedes the response calculation. The results below do not apply an operator-norm bound to arbitrary unbounded continuum stress.

Proposition 2.1 (Exact source completeness)**.** *For all input states,*

$$\Lambda\rho = \Lambda\sigma \implies \operatorname{tr} H\rho = \operatorname{tr} H\sigma \tag{1}$$

if and only if $H = \Lambda^* O$ for a Hermitian output observable O. For such an O,

$$|\operatorname{tr} H(\rho - \sigma)| \le \inf_{a\in\mathbb{R}} \|O - aI\|_\infty \|\Lambda\rho - \Lambda\sigma\|_1. \tag{2}$$

Proof. A Hermitian $D \in \ker\Lambda$ has trace zero. For sufficiently small $t > 0$, $I/d \pm tD$ are states. The implication thus gives $\operatorname{tr} HD = 0$. Finite-dimensional duality identifies the orthogonal complement of the kernel with $\operatorname{ran}\Lambda^*$. Conversely, $H = \Lambda^* O$ makes its expectation constant on every output fibre. The final inequality is trace/operator-norm duality applied to the trace-zero output difference [1]. □

Exact source completeness does not require full state tomography. When it fails, the missing source has a sharp quantitative size:

$$\delta_H(\Lambda) = \inf_{O=O^*} \|H - \Lambda^* O\|_\infty. \tag{3}$$

Theorem 2.2 (Hidden-source size and minimax error)**.** *The infimum in* (3) *is attained, and*

$$\delta_H(\Lambda) = \frac{1}{2} \max_{\Lambda\rho=\Lambda\sigma} |\operatorname{tr} H(\rho - \sigma)| = \inf_f \sup_\rho |\operatorname{tr} H\rho - f(\Lambda\rho)|. \tag{4}$$

Here f may be any real-valued, nonlinear function of the exact output state. Thus no such function improves the worst-case absolute error of an optimal affine estimate $f(\tau) = \operatorname{tr} O\tau$.

Proof. Work in the real vector space of Hermitian matrices. The range of Λ^* is a finite-dimensional closed subspace, so a best operator-norm approximation to H exists. Norm duality and the Hahn–Banach distance formula give

$$\delta_H(\Lambda) = \max\{\operatorname{tr} HD : D = D^*,\ \Lambda D = 0,\ \|D\|_1 \le 1\}. \tag{5}$$

The kernel is invariant under sign reversal, so the maximum could also use an absolute value. Write $D = D_+ - D_-$ and $c = \operatorname{tr} D_+ = \operatorname{tr} D_- \le 1/2$. For any state τ, the matrices

$$\rho = 2D_+ + (1-2c)\tau, \qquad \sigma = 2D_- + (1-2c)\tau \tag{6}$$

are states with the same output and $\rho - \sigma = 2D$. Conversely every equal-output state pair gives an admissible $D = (\rho - \sigma)/2$. This proves the first equality in (4), with attainment.

On a common output an estimate f gives the same number for both states. At least one of their absolute errors is at least half their source difference. This proves the minimax lower bound. A minimizing O in (3) gives error at most δ_H on every state, proving the upper bound. Trace preservation gives $\Lambda^* I = I$, so the approximation space includes constant offsets as well as the nonconstant observable components. No missing centering assumption is used. □

The proof uses standard quotient-norm duality; its consequence here is a sharp source uncertainty for a specified physical interface. The theorem concerns information in an *exact output state*. It does not give a finite observer that state for free. It also does not grant a measurement of an arbitrary element of $\operatorname{ran}\Lambda^*$. An actual instrument is needed for the next step.

Proposition 2.3 (Finite-instrument source interval)**.** *Suppose an allowed output POVM has effects F_y and real scores c_y, with $O = \sum_y c_y F_y$, score width $w = \max_y c_y - \min_y c_y$, and $\delta_O = \|H - \Lambda^* O\|_\infty$. Use this same instrument on n independently acquired outputs of the same input state. With probability at least $1-\zeta$, the sample mean $\bar{c}$ satisfies*

$$|\operatorname{tr} H\rho - \bar{c}| \le \delta_O + w\sqrt{\frac{\log(2/\zeta)}{2n}}. \tag{7}$$

Proof. The output score has mean $\operatorname{tr} O\Lambda\rho$. Its bias relative to the source is at most δ_O. Hoeffding's bounded-variable inequality [2] bounds the deviation of its sample mean by the second term: the centered score moment-generating function is at most $\exp(\lambda^2 w^2/8)$, and the independent product followed by the Chernoff bound gives $2\exp(-2nt^2/w^2)$ for a two-sided deviation t. Adding the bias proves (7); for $w = 0$ the sampling term vanishes directly. □

The interval is conditional on the source model, the actual effects and their stable calibration. Copy preparation, readout, storage and the independence of the acquired outputs are inputs, not consequences of dual-range membership. Independently producing the copies is not cloning an arbitrary unknown state. Existing references and spent ports remain part of the physical ledger, even if their measurement is outside the declared permission. The certificate may be used in a supplied response law: if a response is L-Lipschitz in this source datum, its corresponding error is bounded by L times (7). That step does not derive the response law itself.

3 A hidden interaction, then its actual completion

The distinction can be realized on one finite source rather than by comparing unrelated channels. Take two source qubits A, B, with

$$H = E_0 I + \omega P_{1,A} + g Z_A Z_B, \qquad \omega \ge 0, \quad E_0 \ge |g|. \tag{8}$$

This is a positive energy observable, including the interaction. If only A is available, $\Lambda = \operatorname{tr}_B$ and the output observable $O_A = E_0 I + \omega P_{1,A}$ gives residual norm $|g|$. The two states $|00\rangle$ and $|01\rangle$ have the same A output and energy difference $2g$. Theorem 2.2 therefore gives

$$\delta_H(\operatorname{tr}_B) = |g|. \tag{9}$$

Even complete knowledge of the A state leaves this error floor. It is not a lack of data analysis, and it does not remove gZ_AZ_B from H.

For an actual record use the partially mixed ports of HAFF E and F [5, 6]. A fresh port initially in $\eta_v = (I + vZ)/2$, $0 < v \leq 1$, is coupled by controlled X from its source qubit and read in the port's Z basis. With source eigenvalue $z = \pm 1$, its output obeys

$$\Pr(Y = y \mid z) = \frac{1 + yvz}{2}, \qquad F_y = \frac{I + yvZ}{2}. \tag{10}$$

The coupling and subsequent readout give this law on the actual joint state. They dephase source coherence in the measured basis; they are not the Lüders instrument of the unsharp effect. The source populations, and therefore the expectation of the diagonal Hamiltonian (8), are unchanged.

With only the A port, score the record by

$$c_A(y) = E_0 + \frac{\omega}{2}(1 - y/v). \tag{11}$$

Its mean is $\operatorname{tr} O_A\rho_A$, its width is ω/v, and the residual bias is bounded by $|g|$. Thus the same hidden-source floor is accompanied by an attainable finite certificate, not just an abstract optimal observable.

Now add a second independent port coupled to B on each *same* source pair. The physical joint effects are $F_{y_A} \otimes F_{y_B}$; the outcomes are independent conditional on the two source labels, not independent after averaging over an arbitrary correlated source. Score the paired record by

$$c(y_A, y_B) = E_0 + \frac{\omega}{2}(1 - y_A/v) + g\frac{y_A y_B}{v^2}. \tag{12}$$

Directly summing the four effects with these scores gives H. The interaction bias is now zero. A score-width bound is $w \leq \omega/v + 2|g|/v^2$, so Proposition 2.3 gives a finite full-source certificate on every two-qubit state. The extra port has supplied the missing relation Z_AZ_B, not merely refined the old A marginal.

For $E_0 = 1$, $\omega = 2$, $g = 1/4$, $v = 0.6$, a 95% certificate of absolute error at most 0.5 is guaranteed by 328 independent source pairs and 328 ports in the one-port protocol, or by 165 source pairs and 330 ports in the completed protocol. These conservative budgets use (7). They show a tradeoff between source acquisition and port stock, not a universal saving in total energy. Each port brings purity deficit $\log 2 - h((1 + v)/2)$, where h is binary entropy. The known v, controlled gates, independent production, readout, paired labels and retained archive all count. No thermal reset or perpetual copy supply has been implemented by this example.

The source-completion problem is therefore constructive: identify a hidden source direction, add a physical read of the missing relation, and recalculate both the residual and the statistical budget. The algebraic criterion alone would miss the costs and gains of that change.

4 Algebraic completeness can still be unusable

The opposite limit matters as well. Let a qubit dephasing channel obey $\Lambda_\eta(I, X, Y, Z) = (I, \eta X, \eta Y, Z)$, $0 \leq \eta \leq 1$, and take the positive observable $H = E(I + X)$, $E > 0$. For $\eta > 0$ the dual range is full, $\delta_H = 0$ and $O = E(I + X/\eta)$ reconstructs the energy. The coefficient in (2) is E/η. At $\eta = 0$, the states $\rho_\pm = (I \pm X)/2$ have identical outputs and energy difference $2E$, so $\delta_H = E$.

For $0 < \eta < 1$ these two outputs commute. Their relative entropy is

$$d_\eta = \eta \log \frac{1+\eta}{1-\eta} = 2\eta^2 + O(\eta^4). \tag{13}$$

With n independent copies it is nd_η. The standard Bretagnolle–Huber testing bound [3] gives $P_e \geq \frac{1}{4}e^{-nd_\eta}$ for equal prior probabilities. For the commuting distributions, Jensen bounds their squared Hellinger affinity below by e^{-nd_η}, and Cauchy–Schwarz bounds total variation above by $\sqrt{1-e^{-nd_\eta}}$, giving this testing bound. Thus $P_e \leq \varepsilon < 1/4$ requires $n \geq \log(1/(4\varepsilon))/d_\eta$. A joint quantum measurement cannot evade the classical bound for these commuting states. Measuring X attains an actual informative binary experiment if that measurement is permitted; an additional noisy port introduces its own gain and cost. Small output gain can make source resolution expensive long before the exact range loses a direction. This is a finite-source example, not a model of an absolute vacuum-density shift.

5 The vacuum curvature test

5.1 Null data miss a physical trace

Lemma 5.1 (Complete null-stress kernel)**.** *In a Lorentzian vector space of dimension $D \geq 2$, a symmetric tensor S has $S_{\mu\nu}k^\mu k^\nu = 0$ for every null k if and only if $S = \alpha g$.*

Proof. In an orthonormal frame set $k = (1, \mathbf{n})$, $|\mathbf{n}| = 1$. Comparing $\mathbf{n}$ and $-\mathbf{n}$ gives $S_{0i} = 0$. The spatial quadratic form is constant on the unit sphere, so $S_{ij} = -S_{00}\delta_{ij}$. Hence $S = -S_{00}g$. The two null directions give the same result for $D = 2$; the converse is immediate. □

A physical vacuum-density difference $-\delta\rho\, g_{\mu\nu}$ lies in this kernel. Null tomography, even ideal and complete in its own menu, cannot certify that source component. If two smooth stress tensors are separately conserved on a connected domain, their blind difference $\alpha(x)g_{\mu\nu}$ must satisfy $\nabla^\mu(\alpha g_{\mu\nu}) = \partial_\nu\alpha = 0$. The conserved ambiguity is therefore a constant metric-trace term, not an arbitrary position-dependent vacuum profile. The closed task in the null-cluster paper [21] illustrates both sides: null records estimate a gradient contribution efficiently, yet remain exactly blind to a potential-energy difference. In HAFF D's supplied JT response, the null components of a constant vacuum shift vanish while the dilaton shifts by $\kappa L^2\delta\rho$ [4]. Such a shift gives a response uncertainty if its source is unmeasured; it is not evidence for degravitation. A trace-sensitive datum or independently fixed parameter is required.

5.2 A convention is not a changed stress tensor

The semiclassical source equation is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\langle T_{\mu\nu}\rangle_{\text{ren}}. \tag{14}$$

A Lorentz-invariant vacuum contribution has the form $\langle T_{\mu\nu}\rangle_{\text{vac}} = -\rho_{\text{vac}}g_{\mu\nu}$. The renormalized parameter and its common calibration matter; a bare regulator sum $\frac{1}{2}\sum_{\mathbf{k}}\hbar\omega_{\mathbf{k}}$ is not a parameter-free prediction of that stress [7, 9].

Two changes called "constant shifts" must be separated. In ordinary quantum mechanics, $H \mapsto H + CI$ with the stress tensor held fixed changes only an overall phase. A real vacuum-density difference changes the stress tensor and enters the semiclassical source equation. The

first is nonphysical bookkeeping; the second remains physical even if a chosen interface cannot see it. The distinction is standard, not a new cosmological-constant cancellation mechanism.

The appropriate operational rule is therefore a *source-completion principle*: before predicting a response from restricted data, either certify the needed source observables or retain their residual uncertainty. For finite sectors, Theorem 2.2 gives the sharp residual floor and Proposition 2.3 connects it to records. This rule does not prescribe a new gravitational coupling. A stronger law that filters a physical source would require its own action, conservation and backreaction analysis; philosophical closure cannot supply those equations.

5.3 One material, one calibration

The local Gaussian material in HAFF D supplies a positive comparison [4]. With a fixed texture, masses and boundaries, its active spatial form contains

$$q_a(\phi,\Theta)=\int\big(|\phi'+a(x)\Theta|^2+|\Theta'|^2+m^2|\Theta|^2\big)dx,\qquad m>0. \tag{15}$$

On a finite Dirichlet interval, with a smooth and compactly supported away from its boundary, the matched operator is

$$K_a=\begin{pmatrix}-\partial^2 & -\partial\circ a\\ a\partial & -\partial^2+a^2+m^2\end{pmatrix}. \tag{16}$$

The off-diagonal entries are adjoints. The positive squares establish stability. Reducing to a minimal operator gives a potential-trace difference $a^2/2$; the standard heat coefficient [11] then yields

$$\Delta E_\Lambda=\frac{1}{8\pi}\int a(x)^2dx\,\log(\Lambda/\mu)+C_\mu[a]+o(1). \tag{17}$$

Matched boundaries cancel their coefficient differences. HAFF D gives the reduction and the same material action's one-loop counterterm; the logarithm is not a universal finite gravitational cost.

Changing a changes the coupled fluctuation operator and a calibrated spectrum or force response. Once material couplings and the finite normalization are fixed by a common comparison, the response cannot be removed by choosing a separate energy zero for each configuration. An unchanged profile with a new Hamiltonian label is still bookkeeping. An interface that misses the profile-sensitive observable has a source-certification problem, not a different physical material.

The counterterm and sensor must be included when varying a common renormalized action. Under a covariant material-functional and state prescription, its Ward identity includes material equations and any compensating dilaton force [4]. A static Euclidean or in–out vacuum comparison does not implement causal control. A controller that changes a requires work, retained records and an initial-state retarded response. Neither (17) nor a visibility label already supplies that controller.

6 Visibility without an eligibility shortcut

Useful evidence for a vacuum comparison includes causal accessibility, boundary conditions, state restriction or modular data, actual probe response, and independently specified asymptotic or sector labels. These are a menu of tests to be fixed *before* calculating the response, not

a five-item law derived from Q-RAIF A. A change in one may provide evidence of a physical difference. An unchanged restricted output establishes indistinguishability in that menu; it cannot classify all possible underlying differences as energy conventions.

Casimir configurations change boundaries, modes and stress response [12]; phase transitions can change spectra, condensates and renormalized vacuum parameters. A trace anomaly is a physical renormalized stress effect rather than a phase relabeling [10]. Each needs a matched source calculation. Their possible gravitational significance does not depend on passing an observer's preferred visibility test. Conversely, a counterterm or bare zero-point sum cannot be assessed independently of the common renormalization condition that fixes the physical stress.

This locates the difference from proposed modifications of gravity. Unimodular or trace-free formulations change the role of the trace equation; sequestering introduces constraints on vacuum contributions; degravitation changes the long-wavelength response [17, 18, 19]. The present source certificate does none of these. It bounds what data determine about a specified source, before that source is inserted into a specified response law. Induced-gravity approaches use matter response to construct dynamics and must still match the source and its calibration.

The vacuum-curvature test is consequently two-part. First distinguish a nonphysical offset from a real changed stress using a common material model and calibration. Then ask whether the declared interface certifies that stress well enough to discriminate the candidate response laws. Here source eligibility precedes a numerical response estimate: it means establishing a physical, commonly calibrated source difference, not declaring an unobserved source ineligible to gravitate. A physical vacuum shift in a null blind direction fails the second part until the interface is completed. Computing curvature and using its change as the original visibility datum would make the test circular.

This does not explain why physical Higgs, QCD or phase-transition vacuum contributions produce no curvature at a naive large scale. That remains the cosmological-constant problem. The source certificate prevents a spurious solution by access terminology and identifies what a genuine source-filtering proposal must measure and predict.

7 Closure and the location of the reader

The certificate itself assumes a source observable, a readout channel and an instrument. Can a more complete theory generate them? The question has content even if all three can be represented mathematically. A stored description of an instrument is not its physical execution, and specifying a channel is not a derivation of the body, coupling and records that implement it. An internal model can successfully calibrate some of these components without thereby explaining the origin of every supplied structure.

Conjecture 7.1 (Readout closure)**.** A complete physical account of an operational spacetime must specify or dynamically generate the interfaces that select its records, sectors and source data, including the physical resources that keep them usable. An interface may not be fully internalizable within the description it makes available without an additional level of structure.

The first sentence states the explanatory demand; the second is the stronger possibility to be tested. A typed fixed point, stratified description or dynamically formed executor could meet that demand. Finite programming obstructions and the hidden-source theorem do not rule those routes out. Nor does an unobserved operator imply that no observer-independent world exists. The conjecture concerns whether the completed description accounts for the conditions under which its own distinctions can be physically made.

Multi-observer descriptions impose record-consistency assumptions [13, 14]; island calculations assign radiation and interior sectors [15, 16]; cosmological measures require sampling rules. These are different places to test whether the account includes its physical interface. AdS/CFT, for example, may provide an internal description of bulk access through boundary states and operators. Its success would be evidence for closure, not a reason to redefine the goal.

A useful test must freeze the meaning of the interface before examining a claimed complete construction. The test should ask which structures are primitive, which are generated, how records are physically acquired, and whether source uncertainty and executor costs are accounted for. If a closed theory supplies these objects internally, that is a positive closure result and evidence against the stronger sentence of the conjecture. It cannot be excluded by renaming every successful construction an external readout. Conversely, leaving the necessary instrument as an unpriced oracle does not answer the explanatory demand.

There is a philosophical difference between a world seen incompletely and a world made only by seeing. The hidden-source example insists on the former: two worlds can differ in energy while the observer receives the same record. A new interaction can reveal the difference. The interface both limits knowledge and gives a physical route to improve it. "No view from nowhere" is most useful as a demand to locate that route inside the theory, rather than as a prohibition on seeking an observer-independent account [8].

8 Conclusion

Readout blindness has a quantitative source meaning. In a finite sector, the dual-range distance is exactly the unavoidable error of any estimate from the ideal output state. An actual instrument adds a separate sample budget. The positive two-qubit construction shows how the missing interaction can be recovered through a second priced record of the same source, with the complete source operator reconstructed by the measured joint scores.

Null stress misses the metric trace, including real vacuum-density differences. The matched material comparison shows why changing a coupling cannot be treated as independently resetting an energy zero. Both results require physical source completion before a restricted interface can support a curvature prediction. They do not justify a law in which whatever a reader misses ceases to gravitate.

The closure conjecture remains broader than these theorems. It asks how the interfaces that certify and act on sources come into being, and whether a description can account for its own executor. Concrete source certificates and completion protocols sharpen that question without closing it by definition. A theory that genuinely generates its records, source map and usable apparatus would advance the programme even if it overturned the conjecture's strongest form.

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